

Advanced internetwork design — today and in the future

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This paper highlights the current approach to large-scale internetwork design using today's network layer routing protocols. It describes how large internetworks can be constructed to provide stability, resilience and scalability, together with a set of best practice design guidelines for those involved in the design of large data networks, highlighting the key issues to be considered. The paper looks at how new technologies are likely to enhance this design approach, in particular, how the full potential of cell switching can be realised through emerging technologies such as integrated switching and routing. The paper concludes that the model proposed for large-scale networks based on current routing protocols and technologies can be enhanced to make optimum use of switched WAN services such as ATM, through the use of newer routing technologies.

1. Introduction

The art of large-scale internetwork (data network) design rests in striking the right balance between the promise of feature-rich new technologies and the assurance of well-proven existing technologies to provide resilient network designs. Many of the concepts and design criteria are equally applicable to the global 'Internet' which offers end-to-end connectivity for millions of users. A key component in the design of any intranet or the Internet (referred to hereafter as internetwork) is the network layer 'routing protocol' which makes routing decisions on how data should flow across the network and allows networks to be built which are resilient to link or node failure.

This paper describes the operation of today's network layer routing protocols in terms of traditional internetworks. It goes on to highlight the constraints of such protocols when operating over large switched WAN services such as ATM. The paper describes how it is possible to engineer successful internetworks, working within these constraints, but that in the longer term, enhancements to current network layer routing protocols will enable optimal use of BT's current and proposed switched WAN services. The paper also details a number of best practice design guidelines for the use of today's routing protocols over switched network services.

The paper is limited to the current approach to large-scale internetwork design using the most widely used protocol suite TCP/IP and one of the most common desktop applications based on TCP/IP — Windows NT. The challenges facing the network designer in meeting the key criteria of scalability, resilience and stability are presented

together with the most appropriate technologies and techniques to achieve these goals. Finally, newer technologies in the form of integrated switching and routing are introduced together with the advantages they bring in combining IP and ATM technologies.

2. Today's approach to large-scale internetwork design

Internetwork design is easy — that is until differing user requirements are taken into account. High levels of resilience can add significant complexity to the network layer routing strategy for a design; careful tuning of the routing protocols is necessary to detect failures, select the best alternative path and continue forwarding traffic without time-out of the end-user application. Highly secure networks, such as for financial or government institutions, can alter the whole shape of the design based on the capabilities of network encryption units or centralised authentication servers.

Perhaps the single most important factor influencing the complexity of an internetwork design is scalability which is one of the key themes throughout this paper.

2.1 Traditional internetworks

When the IP network layer protocol and its associated routing protocols were developed the primary wide area network (WAN) technology was the private circuit. Many of the features of the protocols were therefore optimised based upon point-to-point WAN links. This has led to a hierarchical model which is based upon various levels of

both partial and full meshing of network components. A three-tier hierarchical network is depicted in Fig 1.

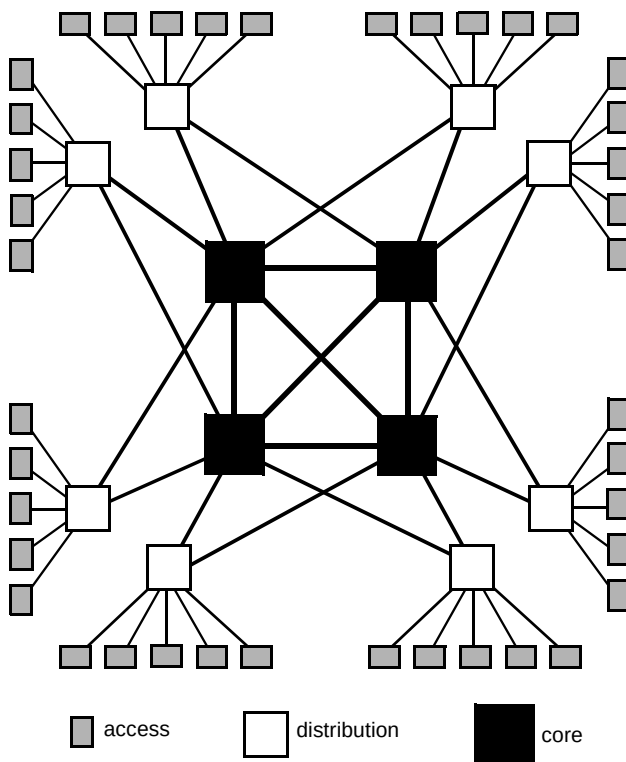


Fig 1 Current hierarchical routing structure.

The model uses the concept of breaking the complete internetwork into separately controlled segments at the routing level. It also attempts to keep local traffic between 'access' nodes parented off the same 'distribution' level to that distribution router. At each stage from access through distribution and finally to core, the level of routing information is being summarised to limit the overhead of maintaining tables and operating the dynamic routing protocols over relatively expensive WAN links. The model is restrictive in that data can never flow directly between any two access routers. It must always traverse multi-hops over the core. A second impact of using this model is the requirement to carefully (and accurately) scale the distribution and core layers, to meet the demands of the access layer.

BT's already successful ATM service, CellStream [1], uses permanent virtual circuits (PVCs) to build scalable, high-performance internetworks. The ideal architecture to replace that shown in Fig 1 is where any access router can communicate directly with any other access router, without going through any intermediate core or distribution routers. CellStream could therefore be further enhanced by the use of switched virtual circuits (SVCs), which offer the promise of such optimal connectivity. However, the existing IP architecture restricts such operation.

2.2 The IP architecture

The IP architecture assumes that each data link subnetwork is labelled with a single IP subnet number [2]. A pair of hosts with the same subnet number communicate directly (i.e. not via routers); a pair of hosts with different subnet numbers always communicate through one or more routers. An IP internetwork is assumed to be composed of hosts and routers interconnected via links (subnetworks).

The 'local/remote' decision is key in IP routing. Each node forwarding an IP packet decides whether the destination is local (attached to the same subnet) or remote (attached to a remote subnet).

If the outcome is 'local', then the host resolves the IP address to a link-layer address (e.g. by using the address resolution protocol (ARP)), and then sends the packet directly to that destination (using the link-layer services). If the outcome is 'remote', then the host uses one of its first-hop routers (thus relying on the services provided by IP routing), forwarding the datagram to this device to be processed and routed to the destination.

To summarise, two of the important attributes of the IP subnet model are:

- hosts with a common subnet address prefix are assumed to be attached to a common link (subnetwork), and thus communicate with each other directly, without any routers — 'local',
- hosts with different subnet address prefixes are assumed to be attached to different links (subnetworks), and thus communicate with each other only through routers — 'remote'.

2.3 Today's routing protocols

The process of building routing tables can be based upon either a manual, static configuration approach or dynamically created based upon algorithms operating within the network's routing nodes.

- Static routing

This requires manual intervention to decide how packets are to be forwarded. It cannot accommodate rapid growth or rapid change. The required addition or deletion of a single router requires all routers to be manually updated.

- Dynamic routing

This gives automatic self-learning and propagation of available paths/routes throughout the network. It is able to cope with network change automatically and requires little manual intervention. However, it

requires additional processing within routers and a proportion of network bandwidth to carry routing protocol traffic. This is the only option if the network is required to provide resilience to link and router failure.

The following discussion focuses on the dynamic routing case.

2.4 Current routing model/hierarchy

The model used when designing large internetworks is shown in Fig 2. It displays a concept where the internetwork is broken into separate smaller entities (domains) which are linked to form a single homogeneous network. The model makes best use of today's routing protocols by limiting overhead on both router resources (CPU and memory) and the network bandwidth required to operate the routing protocol. Typically, there are two types of routing protocol — intra-domain protocols operate within a domain and inter-domain protocols operate between domains.

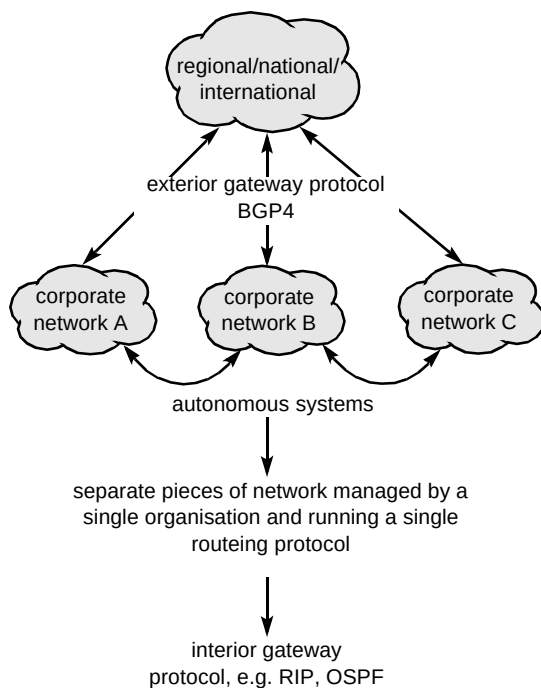


Fig 2 Dynamic routing model.

Intra-domain routing

Intra-domain (interior) routing is intended to perform two main tasks:

- find the 'optimal' path (according to a given metric and/or policy) across the routing domain for any given destination,
- adapt to router or link failure within the domain as quickly as possible and to minimise the disruption caused by such a failure.

Intra-domain routing describes a mechanism to route datagrams through a single network domain (e.g. a corporate network). The dynamic routing protocols used within intra-domain routing use either a flat hierarchy (e.g. RIP), or require the creation of an explicit hierarchical topology, through the construction of a backbone and logical routing areas. OSPF and ISIS are examples of protocols which require this latter type of configuration.

The open shortest path first (OSPF) protocol

OSPF [3] was developed in 1991 and was defined by an Internet Engineering Task Force (IETF) working group. It describes an intra-domain routing protocol for IP and provides hierarchical routing, which is described below, through the use of a link state-based routing algorithm.

OSPF forces a two-level hierarchy on the routing domain — a backbone 'area' and a number of separate routing areas (leaf areas) connected to it (see Fig 3). In this context, an area is simply a subset of the network which contains interconnected routers. The 'backbone' area is connected to the other areas by special routers which are part of the backbone and a 'leaf' area. This structure imposes hierarchy and hence routing control within a routing domain. It is restrictive in that traffic from a node in area 1 can only get to area 2 by passing into the backbone and then into area 2. However, in the traditional mesh case of a 'core-distribution-access' topology network (Fig 1), this maps well to the underlying mesh network topology.

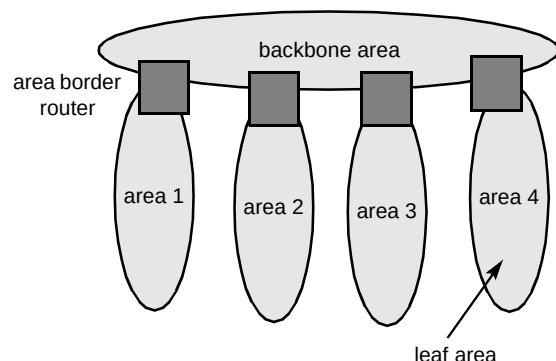


Fig 3 OSPF two-level hierarchy.

OSPF uses three basic components in its operation:

- a 'hello' protocol to establish and maintain neighbour relationships,
- a method of exchanging and disseminating routing information,
- an algorithm to determine the least cost/shortest set of paths throughout the network to all destinations.

OSPF offers fast convergence and low overhead of routing protocol traffic, maximising the WAN bandwidth

made available to user data traffic; these factors account for its widespread deployment.

Inter-domain routeing

Inter-domain (exterior) routeing protocols differ from intra-domain routeing protocols due to a differing set of operational requirements. Although inter-domain routeing protocols may provide some mechanism for 'optimal' path selection and for rerouteing on link or node failure, they are more concerned with:

- exchanging 'reachability' information for very large numbers of destinations between domains,
- enforcing 'policy' (predefined rules) on the routeing paths taken across the network.

The main requirements for inter-domain routeing protocols come from the global Internet. The Internet has a very large number of separate routeing domains which are interconnected by a single inter-domain routeing protocol. This protocol has to handle advertising reachability to all possible destinations on the Internet. (With various address aggregation and summarisation techniques, this stands at approximately 34 000 routeing entries at the time of writing.) It also has to enforce routeing policy decisions made by different Internet service providers. Above all, inter-domain routeing protocols must be highly stable and predictable in operation.

The border gateway protocol (BGP)

There is one primary inter-domain routeing protocol used in the core of the Internet today — the border gateway protocol version 4 (BGP4) [4]. This provides simple mechanisms for choosing optimal paths to destinations and for recovering from router or link failure, but very sophisticated mechanisms for advertising large quantities of 'reachability' information and applying routeing policies. It is largely due to the abilities of BGP4, that the Internet has been able to scale to its current size.

The operation of BGP is based on the identification of each separate routeing domain by a unique autonomous system (AS) number. For the Internet, these are allocated by a centralised addressing body. BGP assumes that an intra-domain routeing protocol is run within each routeing domain, and that it is given the output of the intra-domain routeing protocol (i.e. addresses of all destinations reachable within the domain) for advertising outside the domain. BGP also injects reachability information for destinations outside the routeing domain into the local intra-domain routeing protocol.

A BGP router holds three databases:

- routes that each of its peers have advertised to it,
- its local routeing table,
- routes that it will advertise to its peers.

When a BGP router advertises that it can reach destinations to a peer router, it includes the AS number of the routeing domain that contains the destination, and the AS numbers of the routeing domains that need to be traversed to get to that destination. This allows the BGP router receiving the information to make a decision based on various configured policies.

BGP has attributes of both a traditional link state and a distance vector algorithm. By using TCP as a reliable transmission session between peering routers, information is only sent once and then updated by only sending changes. This is required to allow very large amounts of information to be forwarded between BGP peers in the core of the Internet. Each BGP router makes a decision based on its own policy definitions on which path it would use to get to each destination. It is this chosen path that it advertises to its peers. Hence BGP uses hop-by-hop decision making (like a distance vector protocol), but only sends changed information (like a link-state protocol).

Large BGP networks can take a long period of time (several minutes) to propagate new routeing information across the BGP backbone. This is actually very useful for very large networks such as the Internet as it damps the speed of response to changes in network topology (link failures, etc). This is important as a large network would become unstable and could potentially 'melt-down' if there were very frequent changes in routeing information being advertised between BGP peers. Each BGP router would spend all its time calculating new routeing tables and no time actually forwarding traffic. The protocol takes explicit measures to ensure that unreachable routes are removed from routeing tables more rapidly than new routes are added. This again helps in the stability of very large BGP networks, such as the Internet.

As an example, BGP is normally deployed as a backbone protocol for interconnecting different routeing domains (e.g. different corporate networks). If the mesh network topology of Fig 1 is considered to be a typical routeing domain, then Fig 4 shows how BGP could be used to interconnect them. One or more routers within each AS (domain) are selected as autonomous system boundary routers and these are interconnected by some form of leased line mesh. BGP peers are then configured between some or all of the AS boundary routers.

It is a requirement of BGP that all BGP speaking routers in the same AS must have a peer to each other (i.e. be fully

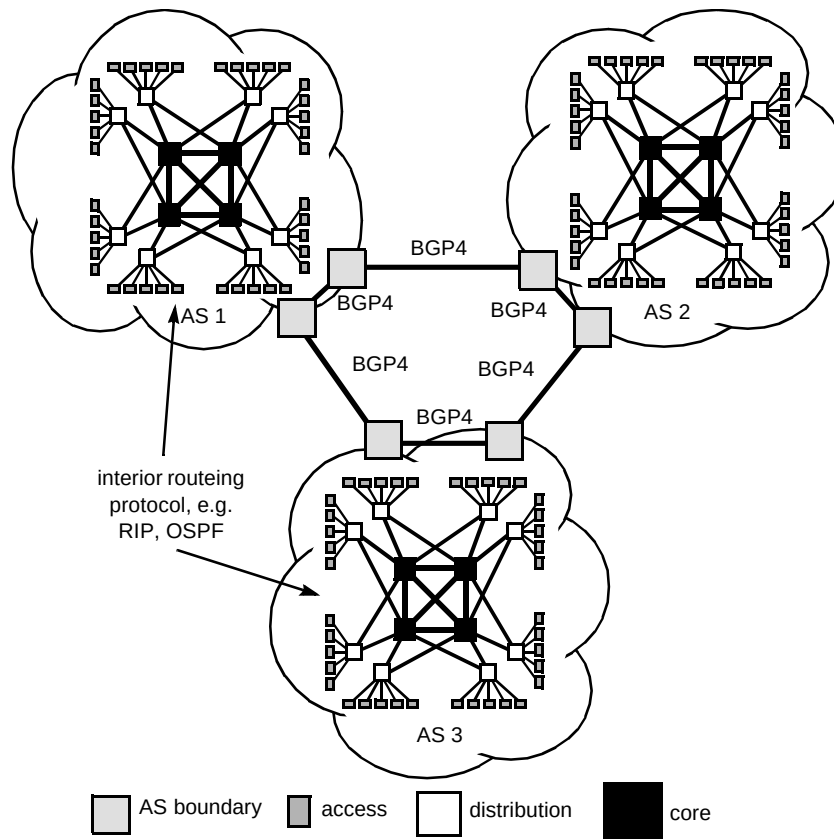


Fig 4 BGP example.

meshed). The peering between routers in different ASs can be either a full or partial mesh. The peering is normally the same as the physical topology of the interconnections between the AS boundary routers. In the case of the diagram, there is a full mesh between ASs, but not a full mesh between all AS boundary routers. In this example, all traffic between ASs follows the same path as the routing information and hence traverses one AS boundary router in each AS.

2.5 Limitations of traditional protocols for 'large cloud networks'

The aim of introducing new WAN services which provide the so-called 'cloud model', as depicted in Fig 5, is to allow the concept of any-to-any connectivity between nodes/routers accessing the cloud. This means that all devices connected to the network can send and receive data directly (in a single hop) from any other network-connected device. In this case the network architecture and especially the network layer routing protocols must scale to support such numbers of nodes/routers and destination networks.

There are two important differences between a 'traditional' internetwork based on meshes of point-to-point connections (for which many of today's protocols are optimised) and an internetwork using a switched WAN service (cloud networks):

- the IP subnet model becomes restrictive for cloud networks,
- the number of neighbours a router can have is significantly increased.

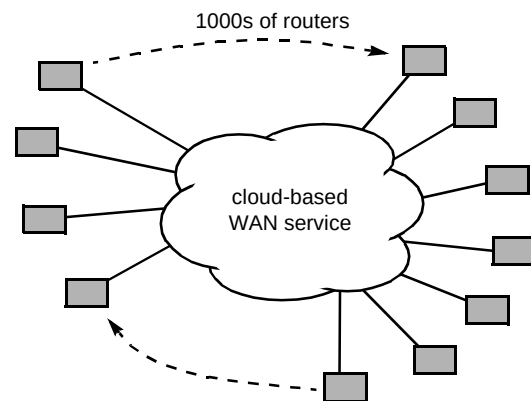


Fig 5 Cloud-based WAN service model.

These issues are highlighted below, followed by the impact when building a network over a cloud network using OSPF or BGP.

2.6 Applying the IP subnet model to ATM SVC networks

Future ATM networks could allow an SVC to be set up on demand between any two end points which are

communicating. However, using the existing IP subnet model would require SVC establishment only between communicating entities that are on a common IP subnetwork. For hosts attached to different logical IP subnets (LIS), SVC establishment would require the set-up from each end-point to its transit router, and as required between the intermediate routers. This is likely to force the majority of traffic between ATM attached hosts (or ATM attached routers) through one or more intermediate router hops, as displayed in Fig 6.

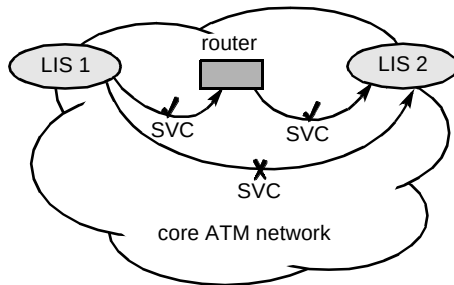


Fig 6 LIS model overlaid on switched WAN.

2.7 Neighbour count issues

In a traditional mesh network, it would be unusual for even a core router to have more than 40 'neighbours', i.e. 40 other routers to which it is directly connected and with which it exchanges routing protocol packets. In the switched WAN service environment, a router could have as many neighbours as there are routers connected to the switched service. For even a moderately large network, this leads to a massive change in the amount of routing protocol traffic that a router would be required to send and receive. If it were assumed that there is no restriction imposed by the IP subnet model, the ideal case would be for all routers connected to the WAN service to be on the same subnet and to communicate directly. Unfortunately, today's routing protocols were not designed for this type of operation and so careful network design is required in this environment.

The problems created by having a large number of neighbours are discussed in the following sections for each of the routing protocols previously described.

2.8 Use of OSPF over large clouds

In essence the concept of the current routing model relates to the creation of logical IP subnetworks (LISs). Unfortunately, it is not possible to build networks using OSPF to form the single flat cloud model of Fig 5. This is because the flat cloud model can only be achieved using a single LIS for the cloud. With OSPF this would also be a single broadcast multiple-access (BMA) network which would all be in one OSPF area (probably area 0, the backbone). This topology is limited by the neighbour count

problem because of the operation of the OSPF's 'hello' and 'database exchange' mechanisms. All routers send periodic hello packets to all other routers in the LIS, and each hello packet carries a list of all neighbours the router has seen. The level of hello traffic received by each router therefore scales as the square of the number of neighbours. The database exchange mechanism used to propagate changes in network state scales as the square or cube of the number of neighbours, depending upon the network event triggering its operation. Thus router performance limits the number of neighbours it is possible to have.

There is also a practical limit to the number of routers in a single OSPF area, as discussed below. This also restricts the single flat cloud model.

OSPF can be used for large-scale networks if the standard OSPF hierarchical model described by Fig 3 is imposed on the cloud network. This breaks up the cloud network into a number of smaller, logically isolated, clouds which cannot communicate with each other directly. Traffic can only get from one cloud to another via 'transit routers' which are connected to both clouds. Figure 7 shows this model for traffic flowing from router X at the edge of the network to router Y also at the edge of the network. As routers X and Y are in different logically isolated clouds, traffic must pass from X to transit router A, then to transit router B and then on to Y. This hierarchical structure can force significant quantities of traffic to flow through the transit routers. In order to maximise the size of each logical cloud in the hierarchy, the transit routers are usually configured to also be in separate OSPF areas. This makes each transit router an OSPF area border router (ABR).

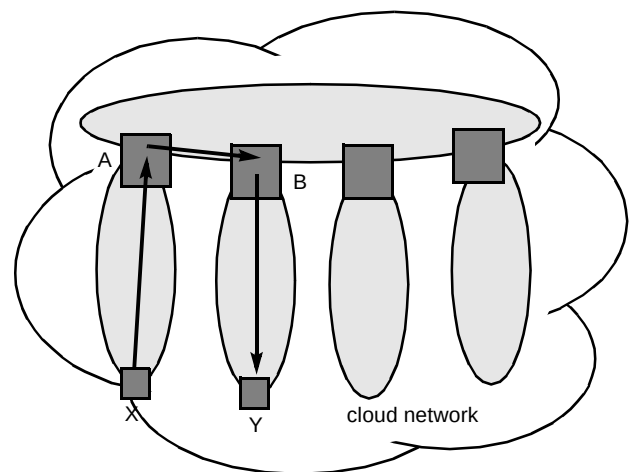


Fig 7 Operation of transit routers.

The following sections discuss some of the scalability issues.

Shortest path first algorithm and database size

By keeping areas small and making effective use of route summarisation, the impact of router CPU usage can be minimised. However, keeping the number of routers per area low increases the number of areas required and hence the number of area border routers. Hence increasing the population of routers within the backbone area. Additionally, routers require memory to maintain both the database and the extrapolated routing table. Again, by maintaining small areas, the size of both the link state database and routing tables is kept small. It should be noted that the backbone area is also subject to the same scaling issues as other areas within OSPF.

Network scalability

Current recommendations suggest that a maximum of 40 to 60 routers should be used within any OSPF area. Additionally, the population of routers within the backbone area should be kept toward the lower value (to maintain as high a degree of stability within area 0 as possible). This model effectively creates the concept of transit routers — any data transfer between hosts within different areas must traverse area border routers and the configured backbone. Key factors to consider with this approach include the additional cost of router hardware (which is usually dedicated to the transit router function), additional network accesses to provide access to the backbone and other transit routers, multi-hop routing and the associated increased transit latency, and the potential for a single point of failure.

Resilience

OSPF makes use of an area border router which provides data transfer between routers within the area and the backbone — effectively a transit router. BT's data customers demand high levels of network availability and resilience. To enhance the model the topology must be modified to include a second area border router connection — as depicted in Fig 8.

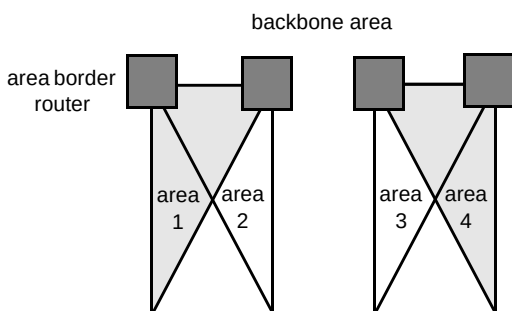


Fig 8 Resilient OSPF topology.

Although this provides increased resilience, trade-offs must be made against increased CPU and memory usage within the ABR transit routers. These generally must be

substituted for higher performance units, now operating within three OSPF areas and maintaining three OSPF link state databases, each operating three instances of the shortest path algorithm.

Speed of convergence versus network overhead

Following network change, the speed of convergence to a new network topology is mainly governed by the time for the router to detect loss of a neighbouring router (often referred to as an 'adjacency'). The adjacencies are maintained through the use of the hello protocol with each router periodically exchanging hello packets. To achieve fast convergence following failure of a network access a short hello interval must be used. Hello packets consume network bandwidth and router CPU cycles. As such, a design trade-off forms between fast convergence and the requirements for ever higher speed WAN accesses, higher performance routers and their associated cost.

Rapid change/route flapping — recalculation

During periods of rapid change the quantity of link state advertisements (LSAs) and associated recalculation of the shortest path first (SPF) algorithm can lead to network failure. OSPF attempts to limit the impact of networks cycling up and down by two actions — the first limits the processing of incoming link state advertisements and the second introduces a hold period before running the SPF algorithm after the receipt of an LSA. The minimum LSA arrival interval causes a typical router to only process one copy of the same LSA per second at default values. For example, the hold or damping period from one major vendor's implementation defaults to 5 sec. Such values have been produced by vendors and operators through long experience of network design, and should only be modified after careful consideration.

2.9 Use of BGP for large clouds

BGP is well suited to interconnecting different routing domains over a large cloud network. It works as well on a large cloud network as on a traditional leased line mesh because it only exchanges information between predefined peers. It is possible to configure peers so that a large cloud network looks like a leased line mesh to BGP. By doing this, the neighbour count issues, that restrict use of intra-domain routing protocols, need not restrict BGP.

The fact that BGP carries 'next hop' information with each route allows a flat cloud network to be overlaid by a partial mesh of BGP peers. Even though one router does not peer with all other BGP routers connected to the cloud network, it learns, via its peer, of routes which are direct to the destination. Thus, although routing information is learnt through a partial mesh of BGP peers, in some

circumstances actual traffic can go direct from source to destination over the cloud network.

The disadvantages of using BGP stem from the fact that it was designed as an inter-domain routing protocol and so does not give the optimal route selection and rapid rerouting on failure that intra-domain protocols provide. Additionally, it is necessary to configure all routers connected to the cloud network to be in different ASs. This is only really feasible if the networks at each site (i.e. connected behind each router) are reasonably large in size. Additionally, registered AS numbers (which are required for connection to the Internet) are a scarce, globally allocated, resource.

Figure 9 shows a number of sites interconnected by a single LIS on a cloud network. Each site is a separate routing domain and hence has a separate AS number (AS1 and AS2 here). There are two AS boundary routers per AS which are operating BGP and are peered to each other. There is a partial mesh of BGP peers between these AS boundary routers. Because this partial mesh is across different ASs, next hop information is forwarded by the routers at the core of the mesh (A, B, C, D). This allows router X to send traffic direct to router Y even though routing information is learnt via routers A and B. This topology scales to very large sizes, limited only by the restrictions of the size of the IP subnet to which all BGP routers are connected. However, the fact that BGP is designed to be a highly stable protocol, and that a very large LIS would require routing information to traverse multiple BGP peers (there is a limit to the number of peers per router), means such a network would converge relatively slowly (similar to convergence of the core of the Internet).

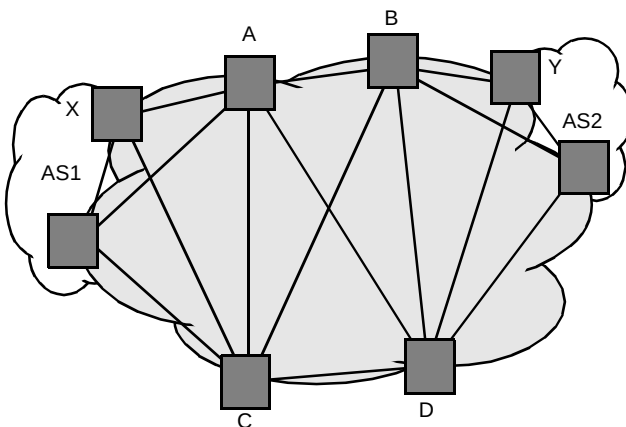


Fig 9 ASs interconnected by BGP.

The growth in the use of large cloud networks seen today is mainly for corporate data networks. These typically have a large number of relatively small sites. The likely deployment of BGP within this type of network is in its normal role as an inter-domain routing protocol which interconnects separate routing domains that run intra-

domain routing protocols. Thus for this configuration a hierarchical approach is required, with only the top layer of the hierarchy using BGP.

2.10 A current solution for large cloud networks

There are a number of solutions using today's routing protocols over cloud technologies. Each is driven by a different set of connectivity and convergence requirements. One possible solution, which has been deployed successfully by BT, is described below.

The model uses multiple OSPF ASs based on the hierarchical model. It also uses BGP to interconnect these routing domains, but does not use a full mesh core, as there are now too many AS boundary routers to have a full mesh of BGP peers. This model would scale well beyond a 45 000 router network. It provides good damping of routing changes to preserve network stability.

Figure 10 shows a network with x ASs, n areas per AS and m routers per area. Practical limits on each of these are also given in Fig 10.

2.11 Intra- and inter-domain routing protocol conclusions

Today's traditional network-layer routing protocols are very scalable. They achieve this predominantly by imposing hierarchy in both addressing and the summarisation of routing information across the network. However, they were optimised for networks based upon leased line meshes. Consequently, in their unmodified form, they restrict the ability to make best use of ATM with SVCs (any-to-any connectivity model) over switched networks because of the imposed hierarchy. Ways in which these limitations can be overcome in the future are described later in section 4.

3. Server/operating system design

The preceding section has described a model for scalable internetwork design. This section looks at how higher layer protocols, in the form of Windows NT, can be supported over such an architecture, exploiting the full power of large corporate intranets by removing geographical barriers.

Working across a company's intranet via e-mail, and other shared resources is second nature to today's average network user. The local file server has evolved into a company-wide resource, made possible through high-performance, stable intranets. This brings with it a number of challenges, e.g. users are typically given an account on each server, which increases administration costs. Many server operating systems have evolved to provide central administration, single user log-in, and improved wide area

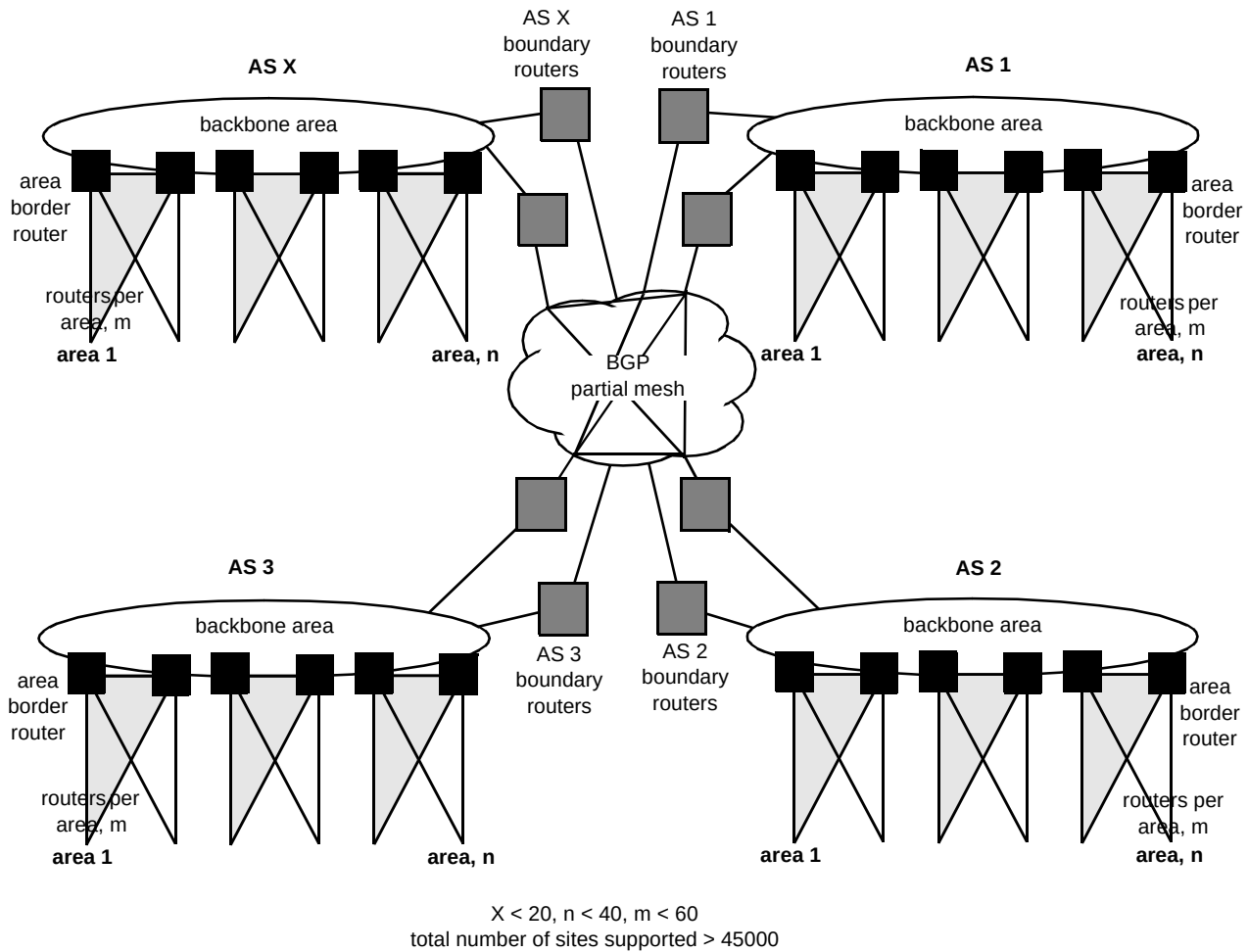


Fig 10 OSPF ASes with a partial mesh BGP core.

support. In particular Novell's NetWare Directory Services (NDS) and Microsoft's Windows NT Domains offer these facilities.

The design of a network is inextricably linked to the server design and in the same way the server design must also be aligned with the network design. The key server-related issues for network designers are:

- providing flexible administration — this may either involve a centralised administration department with delegation of common tasks to local personnel, or provide distributed control to each department within a company,
- providing easy access to resources from anywhere in the company and remote access for teleworking users,
- single log-in — simple for the user, and administrator, with the added benefit of increased security (users tend to keep paper records where multiple passwords cannot be easily remembered),

- minimising the bandwidth required by server background tasks — periodic server-to-server communication maintains an overall 'database' of the servers and resources available, but there is generally a trade-off between overhead on WAN bandwidth and how up to date the network 'database' is.

3.1 Windows NT case study

A model for large-scale NT design, focusing on the key issues identified above, is considered through a case study.

The primary considerations when designing a large-scale Windows NT network are:

- choosing a suitable Microsoft Domain model (usually the Multiple Master Domain Model),
- selecting the right network protocol — TCP/IP is strongly recommended for Windows NT networks,
- minimising the number of domains, while maintaining performance and redundancy,

- grouping users into logical domains, usually geographically, though sometimes by work area (e.g. marketing, sales),
- providing a name resolution system that is simple to administer, reliable, and does not heavily utilise the available bandwidth,
- ensuring rapid convergence of the network 'database' without making large bandwidth demands.

Figure 11 shows a possible layout of the domains, and the location of specific server functions. This diagram overlays on to two of the backbone areas of the underlying networks shown in Fig 10. For this case study, up to 50 routers per area has been assumed, and up to 40 areas per AS. Each branch network contains 10 to 20 users per branch site, resulting in each AS supporting up to 40 000 users.

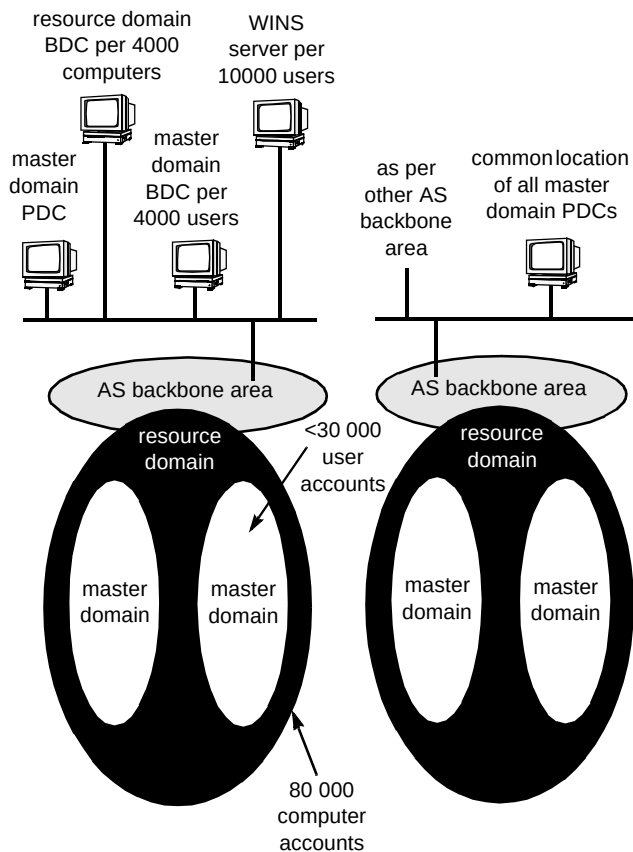


Fig 11 Windows NT design topology.

An NT resource domain only contains computer accounts and can typically handle up to 80 000 computers. A single resource domain has been mapped to each AS.

A master domain is only suitable for up to 30 000 users. As the network consists of up to 40 000 users per AS it is convenient to use two master domains per resource domain.

The size of resource and master domains is limited by the maximum size of account database the domain controllers can handle. The values used here are based on high-performance, dedicated domain controllers with significant quantities of memory.

The logical positioning of the resource domain's controllers (primary domain (PDC) and back-up domain controller (BDC)) and the master domain BDCs is the backbone area of the AS. This means that all authentication is a minimal number of hops away from a branch site. With larger sites and/or a smaller number of sites it may be feasible to position some BDCs nearer the end users; care needs to be taken to ensure that reliable communication between the PDC and BDC is still possible. There should be one BDC per 4000 accounts (this applies to master domain and resource domain). This value is based on how fast the domain controller can respond to requests for authentication by clients. The PDC of each master domain should be collocated on a single backbone area close to the central administrators.

It is recommended that there is one WINS (Windows Internet Naming service — Windows NT utilises WINS to provide name-to-address resolution) server per 10 000 users; this ensures that the server is capable of handling all requests and registrations. The WINS servers will be frequent across all WINS servers in the same AS. One WINS server from each AS will then perform an infrequent replication with one WINS server in each other AS. This minimises replication traffic, provides a reasonable convergence time, and gives a degree of resilience. Some of the WINS servers could be moved nearer to the end users with larger sites and/or a smaller number of sites. Figure 12 shows a WINS replication design for four ASs.

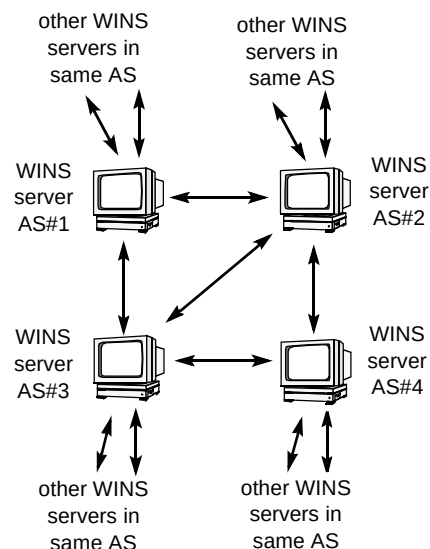


Fig 12 WINS replication design.

(Note that replication times have been deliberately omitted, as this will depend on the desired convergence time.)

The NT browser functionality must be carefully designed such that the most reliable and powerful machine at each branch site becomes the master browser. This will ensure fast browsing at the branch sites, and speed up browser operations over the wide area.

There are many other issues relating to a large-scale NT network design which cannot be covered here in sufficient detail (e.g. tuning of browser traffic), but this insight has shown the importance of a combined server and network design.

4. Future technologies for large internetworks

The two previous sections of this paper have presented the techniques used today to build large internetworks using strict hierarchical topologies. Today's network-layer technology has evolved to operate in this manner and many such networks have been successfully built. However, with the emergence of new layer 2 technologies offering end-to-end switched connections, such as ATM with SVCs, the model used by traditional internetworking solutions becomes somewhat restrictive. Methods to overcome these restrictions are discussed within this section of the paper.

Support of switched circuits using ATM signalling mechanisms, i.e. ATM SVCs, have been defined for several years and are well understood. These mechanisms have been defined as extensions to existing signalling mechanisms from the circuit-switched telecommunications world and provide comprehensive, but relatively heavyweight, mechanisms for establishing end-to-end connections with a specified quality of service. Unfortunately for the existing internetworking community, much of this work assumed that applications within end systems would communicate directly using ATM and hence little initial thought was given to how existing IP networks could make best use of ATM SVCs. More recently, it has become clear that internetworking protocols, particularly IP, will dominate as the data protocols of choice. Hence it has become important for IP networks to be able to make effective use of ATM SVCs in the future.

The majority of the work on supporting internetworking protocols, such as IP, over ATM SVCs over the last three years has concentrated on introducing new internetworking protocols that can make use of the standardised ATM SVC mechanisms. However, over the last year there has been a significant additional development — the introduction of integrated switching and routing approaches from key router and switch vendors. These integrate IP routing into layer 2 switches, e.g. ATM switches, and discard the existing ATM call-control mechanisms. The two main approaches are discussed below.

4.1 Traditional ATM call-control approach

Both the ATM Forum and the IETF have defined methods for supporting internetworking protocols over ATM networks offering traditional ATM SVCs. The IETF has defined 'Classical IP over ATM' [5] and the ATM Forum 'LAN emulation' [6]. Both these approaches define methods for supporting a single sub-network of an IP (or other protocol) network over ATM. They define a server component, to which all client devices attach (via an ATM SVC). This server then informs the client of the correct ATM destination address and the client creates an ATM SVC direct to each destination. However, these mechanisms are not suitable for supporting large internetworks, consisting of many thousands of sub-networks, as a single centralised server approach does not suit this scenario.

An additional protocol has been defined by the IETF for supporting large-scale networks over ATM SVCs. This is called the next hop resolution protocol (NHRP) [7] and has also been used by the ATM Forum as an underlying technology for its multiprotocol over ATM (MPOA) proposal [8]. NHRP allows large internetworks to be built over switched networks by augmenting existing internetworking technologies and protocols, not replacing them. Networks using NHRP will still require the hierarchical topology discussed previously in this paper to be built. Traditional internetworking protocols, such as OSPF and BGP, will be used as the basis for such a topology. This forms the 'routed path' or 'hop-by-hop' path for traffic across the network and is likely to be constructed from ATM PVCs or 'long lived' SVCs. The introduction of NHRP provides a mechanism for specific traffic to bypass this hierarchical topology and make use of an end-to-end switched connection. Such a connection is called a 'cut-through' or 'short-cut'. Figure 13 shows the hop-by-hop and cut-through paths between two routers attached to a large cloud network service offering SVCs, e.g. ATM.

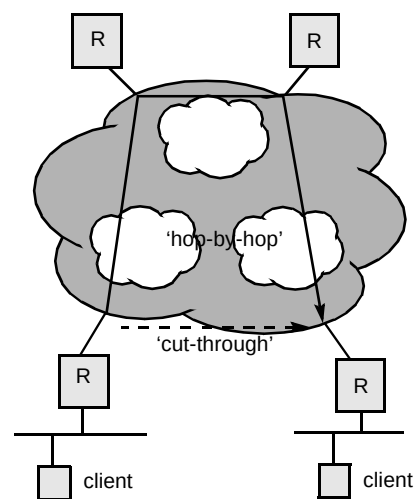


Fig 13 Hop-by-hop and cut-through paths.

NHRP is not a routing mechanism — it relies on the routing protocols used by the hierarchical network to provide routing information for any destination within the internetwork. NHRP provides an enhanced address resolution mechanism. Any device wishing to establish a cut-through connection to a specific network-layer address sends an NHRP request message to the destination across the routed path provided by the hierarchical network. The destination (or a server that responds on its behalf) replies to this message with its ATM layer address in a message which is sent back across the default path. Hence the originator is able to establish an ATM connection direct to the destination.

The scaling properties of NHRP to support very large networks are not yet fully understood as implementations are only just becoming available at the time of writing and minimal deployment has occurred. (In addition, very few wide area ATM networks supporting SVCs exist today.) However, many of the limitations of NHRP are likely to come from the connection-oriented nature of ATM signalling rather than NHRP itself. The ability of NHRP to scale is mainly attributable to the underlying routed network which uses traditional routing protocols such as OSPF and BGP, the scaling properties of which are well known. The major potential limitation of NHRP is the overhead in performing the NHRP and ATM signalling required to establish an end-to-end connection. For many applications, it is unlikely that the overhead required to send NHRP requests and replies, and to signal an ATM connection can be justified and the traffic would be better sent over the hierarchical routed path. Successful use of NHRP is therefore likely to require intelligent systems which only decide to create end-to-end switched connections for application traffic which can justify it, e.g. traffic requiring a specific quality of service and/or of a sufficiently long-lived duration.

The long-term success of this model for supporting large internetworks over ATM will depend on the ability of the ATM network to handle the SVC call volumes (e.g. being able to establish many thousands of SVCs per second) and on the tariff for establishing an SVC. Under such circumstances it becomes possible to support a significant proportion of the internetwork's traffic over SVCs and hence give specific quality of service guarantees to individual applications.

4.2 Integrated switch/router approach

Over the last year an important alternative mechanism for supporting large internetworks over layer 2 switches has emerged — the integrated switch/router approach. This approach has been championed by a handful of vendors from the internetworking community and is currently at the early stages of standardisation within the IETF through the multi-protocol label switching (MPLS) working group [9].

However, it is being driven by some of the major internetworking vendors including Cisco and Cascade and hence appears likely to become an important technology for supporting large internetworks.

The integrated switch router approach attempts to make every node a switch and a router — called a 'label switch router' (LSR) by the IETF. The perceived advantage of this approach is that the network gains the switching performance of, for example, ATM, but retains the control plane used by IP router networks. The same control mechanisms can then be used end to end across the entire IP network. Additionally, using LSRs does not require an overlay network as both the routing and switching functions are integrated into a single layer. Hence the topology of the label switch network is the same as the physical topology. This makes it relatively simple to support using the existing hierarchical topology required by IP protocols, as the physical network is likely to be arranged in a hierarchical manner.

The integrated switch/router approach is depicted in Fig 14; the core of the network uses devices which can route traffic at the network layer and also switch traffic at the ATM layer. The traditional IP network layer routing protocols are used to provide the network with topology discovery and dynamic resilience mechanisms in the same way as a router network. The network may either forward traffic from one router to another using the conventional hop-by-hop mechanism (using both switch and router components of the LSR) labelled 'default-path', or create an ATM-like 'cut-through' connection between the source and destination, which only uses the switch component of the LSR. In this case the source and destination are typically the downstream and upstream integrated switch/router. The protocols used to establish these ATM layer switched connections are currently proprietary to each individual vendor's proposals and are being studied as part of the standardisation effort of the IETF.

The main difference between the different vendor proposals for integrated switching/routing lies mainly in how and when cut-through paths are created. One approach uses detection of user traffic to initiate a cut-through. Another approach creates all possible cut-throughs between all sources and destinations as soon as it has discovered the network topology. These two approaches are designed for different applications — the first is designed to allow cut-throughs for individual traffic flows which require high performance, the second is designed to increase the performance of best-effort traffic within the core of existing large IP networks such as the Internet. As integrated routing and switching evolves it appears likely that these different approaches will begin to merge. This is the intention of the IETF MPLS working group. None of these approaches are currently based on standard technology and, in order to achieve the benefits of this new technology, a

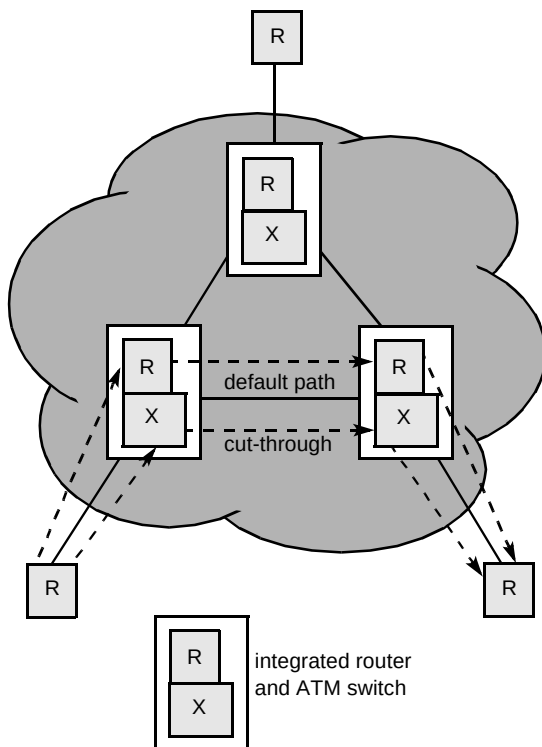


Fig 14 Integrated switch/router network.

standard solution is necessary. In advance of this, it is likely that individual networks will deploy proprietary solutions available from the various vendors. These have the advantage that they leverage the existing IP control protocols and hence can integrate relatively easily with existing IP router networks.

The impact of integrated switch/routers and MPLS on the industry cannot yet be foretold. At one extreme it could become the default mechanism for supporting data traffic over cell switches and hence marginalise ATM. At the other extreme it may remain a niche technology for use in specific situations such as the core of the Internet backbone where existing technologies are experiencing performance problems.

4.3 Conclusions on future technologies for large internetworks

It is impossible to predict the exact technologies which will dominate internetwork solutions over the next few years. However, one thing appears highly likely — Internet protocol (IP) will be the dominant protocol used to transfer application traffic between users. It is therefore vital that future technologies provide good support for IP traffic. Where customers require the support of existing internetworking solutions over ATM networks which provide SVCs, technologies based on the NHRP protocol will be useful. However, this will only become viable on a widespread scale if ATM SVCs become widely available at an attractive cost to customers. Unless this occurs, internetworks making use of ATM SVCs are likely to be limited to high-value corporate networks where the user is

willing to pay the cost associated with guaranteeing quality of service through an SVC. The alternate emerging technology of integrated switches and routers is currently immature, but promises much to existing IP networks as it reuses all of the existing IP control protocols. Currently different vendor proposals attempt to solve different issues arising from the construction of large high-speed internetworks. However, they are likely to evolve to a more generic technology which could offer the benefits of the ATM SVC solution and the traditional router network in one technology.

5. Conclusions

A well-proven approach to engineering large-scale networks based on cloud services has been described using current routing protocols and technologies. Several of BT's large corporate customer networks have been designed by the Broadband and Data Networks teams at BT Laboratories and are currently being or have been deployed using the hierarchical models described within this paper. However, to make optimum use of the capabilities of future switched WAN services, such as ATM with SVCs, new routing technologies and protocols are required. These will still necessitate the application of the hierarchical routing model described within this document, but will enhance the network to allow increased levels of connectivity directly between end hosts.

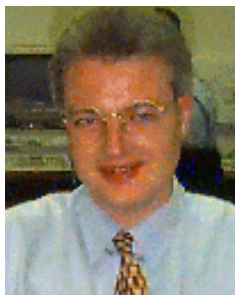
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Paul Hitchen joined BT Laboratories after graduating from the University of Salford in 1990. He became a member of the first BT technical team specialising in the evaluation and deployment of internetworking technology to BT customers. He has wide experience of data networking technology and works closely supporting BT's customer-facing divisions to design and sell network solutions to BT's customers. He currently leads a project for Syncordia Solutions with responsibility for the development of internetwork design guidelines, evaluation of new internetworking technologies and



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Paul Mylotte holds a BSc in Physics from the University of Leeds and an MSc in Telecommunications Systems Design from the University of Essex. He joined BT Laboratories (BTL) in 1973 working on reliability studies of microelectronics. On moving to London he was responsible for the formulation of network plans for quality of service standards for telephony and data networks. He then left BT and worked for GPT for 5 years on the testing and commissioning of System X sub-systems, and returned to BT in 1985 to take up responsibilities for System X software project management and testing. This was followed by 4 years in international standards work covering ISDN and broadband signalling standards, during which time he represented BT at ETSI, ITU-T and ESIG. He is currently working in the Broadband and Data Networks Unit at BTL on various aspects of IP/ATM, and runs the IP/ATM topic group which is responsible for helping to co-ordinate the BT view on the issues. He is a Chartered Engineer and MIEE.